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The analysis of EEG coherence reflects middle childhood differences in mathematical achievement

Andrés A. González-Garrido[a,b,*], Fabiola R. Gómez-Velázquez[a], Ricardo A. Salido-Ruiz[c], Aurora Espinoza-Valdez[c], Hugo Vélez-Pérez[c], Rebeca Romo-Vazquez[c], Geisa B. Gallardo-Moreno[a], Vanessa D. Ruiz-Stovel[a], Alicia Martínez-Ramos[d], Gustavo Berumen[a]

[a] Instituto de Neurociencias, Universidad de Guadalajara, Francisco de Quevedo 180, Col. Arcos Vallarta, Guadalajara, Jalisco 44130, Mexico
[b] O.P.D. Hospital Civil de Guadalajara, Calle Coronel Calderón #777, El Retiro, 44280 Guadalajara, Jalisco, Mexico
[c] Departamento de Ciencias Computacionales, CUCEI, Universidad de Guadalajara, Mexico
[d] Departamento de Neurociencias, CUCS, Universidad de Guadalajara, Mexico

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ABSTRACT

Symbolic numerical magnitude processing is crucial to arithmetic development, and it is thought to be supported by the functional activation of several brain-interconnected structures. In this context, EEG beta oscillations have been recently associated with attention and working memory processing that underlie math achievement. Due to that EEG coherence represents a useful measure of brain functional connectivity, we aimed to contrast the EEG coherence in forty 8-to-9-year-old children with different math skill levels (High: HA, and Low achievement: LA) according to their arithmetic scores in the Fourth Edition of the Wide Range Achievement Test (WRAT-4) while performing a symbolic magnitude comparison task (i.e. determining which of two numbers is numerically larger). The analysis showed significantly greater coherence over the right hemisphere in the two groups, but with a distinctive connectivity pattern. Whereas functional connectivity in the HA group was predominant in parietal areas, especially involving beta frequencies, the LA group showed more extensive frontoparietal relationships, with higher participation of delta, theta and alpha band frequencies, along with a distinct time–frequency domain expression. The results seem to reflect that lower math achievements in children mainly associate with cognitive processing steps beyond stimulus encoding, along with the need of further attentional resources and cognitive control than their peers, suggesting a lower degree of numerical processing automation.

1. Introduction

In an increasingly competitive technological world, underachievement in mathematics can dramatically hinder children's opportunities to succeed, not only in the academic setting but also in social and economic future endeavors (Ritchie and Bates, 2013). There is sufficient evidence supporting a strong relationship between mathematical development and the skills required for symbolic numerical magnitude processing (Durand, Hulme, Larkin, & Snowling, 2005; Holloway and Ansari, 2009; Lyons, Price, Vaessen, Blomert, & Ansari, 2014; Sasanguie, De Smedt, Defever, & Reynvoet, 2012; Vogel, Remark, & Ansari, 2015).

The integration of multiple brain regions during early arithmetic processing has been studied using functional magnetic resonance imaging methods (fMRI: Anderson, 2007; Dehaene, Piazza, Pinel, & Cohen, 2003; Rosenberg-Lee, Lovett, & Anderson, 2009). In this regard, and as part of children's cognitive development, arithmetic problem solving has been related to relevant changes in the hippocampal-neocortical functional connectivity (Cho et al., 2012; Qin et al., 2014).

A recent quantitative meta-analysis of fMRI studies in healthy children below the age of 14 revealed that number tasks (those involving numbers without formal calculation) are mainly associated with the significant participation of the inferior parietal lobule (Brodmann area: BA 40), extending to regions of the inferior parietal sulcus, claustrum and insula (BA 13) in the right hemisphere, whereas calculation tasks implicate structures from both hemispheres (Arsalidou, Pawliw-Levac, Sadeghi, & Pascual-Leone in press). Children with math difficulties though, have a reduced ability to process symbolic representations (Iuculano, Tang, Hall, & Butterworth, 2008; Mussolin, Mejias, & Noël, 2010), and they show an inappropriate task modulation and brain hyperconnectivity (Rosenberg-Lee et al., 2015) while performing simple arithmetic operations.

* Corresponding author at: Instituto de Neurociencias, Universidad de Guadalajara, Francisco de Quevedo 180, Col. Arcos Vallarta, Guadalajara, Jalisco 44130, Mexico.
E-mail address: gonzalezgarrido@gmail.com (A.A. González-Garrido).

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Despite the fact that fluency in processing magnitudes associated with Arabic numerals may be a critical foundational skill for arithmetic, few attempts have been made to assess the integrated cortical functions underlying the earlier steps of this process, particularly through the use of tools with high temporal resolution.

Cortical electrical fluctuations can be measured at low cost and noninvasively with electroencephalographic (EEG) sensors placed on the scalp. EEG is a useful tool for functional connectivity analysis since it provides a direct measure of cortical synaptic activity with high temporal resolution. More specifically, EEG coherence; a measure of the degree of association or coupling of frequency spectra between two different time series, is considered a hallmark of integrated cortical functions (Moazami-Goudarizi, Sarnthein, Michels, Moukhtieva, & Jeanmonod, 2008) that has been applied to evaluate functional connectivity in the human cortex (Gross et al., 2001; Srinivasan, Winter, Ding, & Nunez, 2007). In fact, EEG coherence has been interpreted as a reflection of the functional association between different brain regions (Nunez et al., 1997; Sklar, Hanley, & Simmons, 1972; see Lowe et al., 2016 for an extensive review).

In 2005, Mizuhara and colleagues combined simultaneous fMRI and EEG methods to evaluate a group of young adults while performing a simple arithmetic operation (Mizuhara, Wang, Kobayashi, & Yamaguchi, 2005). They found an increase in phase synchronization in beta frequency across biparietal areas that probably represent a functional link between the left angular gyrus and the right superior parietal gyrus, including several other anterior–posterior regions (right dorsolateral prefrontal cortex, putamen, and right superior temporal gyrus). In brief, phase synchronization was interpreted as a possible function of beta oscillations around parietal sites reflecting attentional control for spatial working memory, which has been associated with math achievement (Cirino, Tolar, Fuchs, & Huston-Warren, 2016; Toll, Kroesbergen, & Van Luit, 2016).

Traditionally related with sensorimotor processing (Hari and Salmelin, 1997; Pfurtscheller and Lopez da Silva, 1999), oscillations in the beta frequency have recently become more broadly interpreted as reflecting functional coupling across large distances (Bibbig, Traub, & Whittington, 2002; Kopell, Ermentrout, Whittington, & Traub, 2000), "readiness", and maintenance of the current brain state (Engel and Fries, 2010), as well as implicated in top-down processing (Spitzer and Haegens, 2017).

Spitzer, Blankenburg (2014) examined the oscillatory signature of abstract numerosity information in human working memory, within and across sensory modalities. They found numerosity-dependent modulations of right prefrontal upper beta (20–30 Hz) that probably reflected the extent to which numerosity was assessed by analog estimation, both between and within presentation modalities, and interpreted their findings as an index of abstract numerosity integration, at a supramodal processing stage in working memory.

Recently, Tschentscher and Hauk (2016) evaluated arithmetic problem-solving in adults and reported a strong beta desynchronization in right frontal, parietal, and temporal regions as a function of executive demands interpreted by the authors as been related to the use of procedural strategies. These findings seem to be in line with those from Gaillard et al. (2009), and Trübutschek et al. (2017), who interpreted the alpha/beta desynchronization as a correlate of conscious perception, and a neural state common to conscious perception and conscious working memory. Other studies have partially associated beta event-related desynchronization (ERD) with cognitive processing (Pesonen, Björnberg, Hämäläinen, & Krause, 2006; Weiss and Rappelsberger, 1998), or with the planning and/or preparation of the motor response (Kaiser, Birbaumer, & Lutzenberger, 2001); while beta event-related synchronization (ERS) has been interpreted as reflecting memory processing (Pesonen et al., 2006; Wisniewski, Mercado, Church, Gramann, & Makeig, 2014).

In light of these antecedents, the present study aimed to evaluate the EEG coherence in children with different math skills levels, while performing a symbolic numerical magnitude comparison task, particularly focusing on changes in the beta frequency band. We hypothesized that different level of arithmetic skills in children might associate with distinct brain EEG coherence patterns, as a result of the degree of specialization that has been developed.

2. Methods

2.1. Participants

The sample included forty 8-to-9-year-old children selected from a pool of 441 third-graders from six public and three private elementary schools. In order to assess their mathematical abilities, all children completed the math sub-section of the Wide Range Achievement Test, 4th edition (WRAT-4; Wilkinson and Robertson, 2006). Participants were equally divided into two groups according to their performance on the WRAT-4: Low Achievement (LA; Mean age = 9.1, SD = 0.62; 10 females) having scores within the lower 15th percentile, and High Achievement (HA; Mean age = 9.1, SD = 0.33; 6 females) belonging to the upper 15th percentile. Children were matched according to their educational environment, such that each LA subject was paired with an HA child from the same school.

2.2. Procedure

A group administration of the WRAT-4 arithmetic sub-test was performed in each elementary school halfway through the school year. From 2-to-3 weeks later, on a separate session preceding the experiment, several additional tests were applied: the abbreviated IQ version of the Wechsler Intelligence Scale for Children, WISC-IV (Block Design, and Vocabulary), the Edinburgh Inventory of Handedness, and a custom-designed questionnaire covering the children's neurodevelopment which was answered by parents. They were all administered at the Neuroscience Institute Lab in a single session, lasting no more than 30 min.

All children were right-handed and had an estimated IQ of 90 or higher [Mean (Standard Deviation): HA = 112.0(15.0); LA = 99.4(7.7); $F_{(1,38)} = 3.79$, $p = 0.027$], according to the local norms of the WISC-IV version. None of the participants had repeated a school grade-level, nor had a history of psychiatric or behavioral disorders. Parents provided written informed consent authorizing their child's participation. The review and approval of the Ethics Committee (ET122009-77) were obtained prior to conducting the study. Participants were then asked to perform a symbolic numerical comparison task with simultaneous EEG recording.

The quantities in the symbolic numerical comparison task were presented in pairs of Arabic numerals with values ranging from 5 to 34. The magnitude of the ratio between the number pairs was controlled in the experimental design, so that half the pairs had a small ratio 1:2 (0.5): 5–10, 7–14, 10–20, 11–22, 13–26, 14–28, 17–34, while the other half, had a greater ratio 3:4 (0.75): 6–8, 9–12, 12–16, 15–20, 18–24, 21–28, and 24–32. The left and right presentation of the numerical value with the greater magnitude was counterbalanced. Children were instructed to determine which of the two numbers represented a higher quantity; and depending, on whether the higher magnitude number was on the left, or on the right side of the screen, they had to respond by pressing a key using the corresponding index finger. Children were comfortably seated at a 60-cm distance of a 17-inch LCD monitor (screen resolution = 1280 × 1024) in which each number pair was displayed. The task comprised a total of 2 blocks (56 stimuli each, lasting about 2 min) with a 3 min resting period between them. The pairs were presented in pseudo-random order within each block, and the order of the blocks was counterbalanced among participants.

Each trial began with the presentation of a fixation cross (400 ms), followed by a pair of numerical values (1000 ms), and then a gray screen (400 ms). Participants were instructed to focus on the fixation

cross to minimize ocular artifacts. The software package (MINDTRACER, Neuronic Systems, Inc.) controlled stimuli presentation and automatically marked behavioral responses.

2.3. Stimuli

A total of 112 pairs of Arabic numerals were shown in 54-point Arial font. These images were displayed on a gray background with yellow and blue print for the right and left numerals, respectively.

2.4. Electrophysiological methods

2.4.1. Recording

Electrophysiological (EEG) activity was recorded from 19 scalp electrode sites (C3, C4, CZ, F3, F4, F7, F8, FZ, Fp1, Fp2, O1, O2, P3, P4, PZ, T3, T4, T5, and T6) using an Electro-Cap (Electrocap Inc., Eaton, Ohio). Electrooculograms (EOGs) were additionally recorded from the outer canthus and infraocular orbital ridge of the right eye using 10-mm diameter gold-disc electrodes. Electrode sites were referenced to linked-earlobes. Inter-electrode impedances were < 5 kilo-ohms at 30 Hz. EEG and EOG signals were amplified at a band-pass of 0.5-to-30 Hz (3dB cutoff points of 6 dB/octave roll-off curves) with a sampling period of 5 ms in a MEDICID-04 system (Neuronic Systems, Inc.). Data was collected in trials of 1000-ms intervals segmented into the first 100 ms –used for baseline correction – and the subsequent 900 ms, for all recorded channels. Single-trial data was stored off-line for averaging and analysis.

2.5. Data analysis

2.5.1. Behavioral measures

Correct, incorrect and non-responses, as well as reaction times, were automatically collected by the software and marked in the EEG as it was being recorded. A mixed-design, repeated measures ANOVA with Group (LA and HA) as the between-subject factor, and Ratio (small and large) as the within-subject factor, was performed to examine correct responses and reaction times during experimental task performance.

2.5.2. Signal processing

EEG time windows lasting 1000 ms – from 100 ms before to 900 ms after stimulus onset – corresponding to a correct behavioral response were selected from each participant's EEG recording (i.e. 20 similar epochs of 1000 ms EEG time windows from each LA and 20 from HA participants). In addition to the visual inspection of the EEG, a Blind Source Separation method was used for artifact and noise canceling based on the methodology presented by Romo et al. (2012; see Barthélemy et al., 2017 for review). Afterward, a 3-layer realistic geometry head model of volume conduction was built with the aim to implement the Surface Laplacian solution through a realistic head geometry model originally described for adults (Deng, Winter, Thorpe, & Srinivasan, 2012). Due to the age of our participants, we decided to tailor the final spatial sampling by adjusting our calculations to the averaged coordinates of the sensor position files settled in the EGI 3-D model for children aged 2–9, and 9–18 years, respectively, in each location in which EEG was recorded in our sample. The realistic head geometry model consisted of an innermost layer brain structure, surrounded by two realistic-geometry structures for the skull and scalp. The spatial filter was applied to all the EEG recordings prior to coherence estimation. Subsequently, the brain connectivity's dynamic was estimated from a connectivity matrix calculated using the magnitude-squared coherence from surface Laplacian EEG (Chorlian, Rangaswamy, & Porjesz, 2009; Nunez et al., 1999; Srinivasan, 2007).

In the present study, coherence was calculated among the 19 electrodes placed on the scalp. Therefore, we obtained a 361-coherence spectra symmetric matrix showing the relationship between all electrode sites within the recorded EEG frequency band (0–30 Hz).

To begin with, 'the connectivity degree' of each electrode (electrode of interest: EOI) was estimated with respect to the others. This was done by averaging the coherence spectrum for each of the rows of the coherence spectra matrix excluding the coherence spectrum of each EOI. Thus, 19 averaged coherence spectra representing the connectivity degree of each electrode were obtained. This procedure was performed for each subject from the LA group and the HA group. Subsequently, 31 columns corresponding to the frequency values from 0 to 30 Hz of the coherence spectra were selected. A one-way-ANOVA with Tukey's honest significant difference (HSD) corrections was performed to assess which of the recording sites showed the most significant differences within a 95% to 99% confidence interval, with respect to the means of the connectivity degree among all the electrodes. The Frequency factor was also tested using a similar procedure for each electrode interconnection.

Another one-way-ANOVA with Tukey's HSD corrections was performed to elucidate which of the electrode inter-connections presented the most significant difference within a 95–99% confidence interval, and they were selected by setting two thresholds of $p < 0.05$ and $p < 0.01$, respectively. The coherence spectra in the diagonal (19) of the coherence spectra matrix were removed along with the lower triangular coherence spectra matrix (171). Consequently, the analysis included only the remaining 171 coherence spectra values.

3. Results

3.1. Behavioral performances

The HA group had a significantly greater number of correct responses ($F_{(1, 38)} = 30.131$, $p < 0.001$, $\eta^2 = 0.442$), and faster reaction times ($F_{(1, 38)} = 14.362$, $p < 0.01$, $\eta^2 = 0.274$) than LA. In addition, a ratio-related difference in the performance was found for the amount of correct responses ($F_{(1, 38)} = 89.732$, $p < 0.001$, $\eta^2 = 0.703$), and for the reaction times ($F_{(1, 38)} = 41.570$, $p < 0.001$, $\eta^2 = 0.442$). Namely, responses were more accurate when evaluating longer ratios. However, no significant interaction was found between group and ratio ($p = 0.11$).

3.2. Electrophysiological results

Table 1 shows the frequencies in which coherence reached the most significant differences between the groups. A noteworthy finding is that the performance of the numeric comparison task elicits significant

Table 1
EEG spectral frequencies in which coherence reached the most significant differences between the groups.

Frequency band	Hz	F-value	Degrees of freedom	*p*	Group comparison
θ	4.7	8.41	6838	0.004	HA > LA
	6.3	20.64	6838	0.001	HA > LA
	7.0	7.75	6838	0.006	HA > LA
α	9.4	8.28	6838	0.004	LA > HA
	10.2	8.86	6838	0.003	LA > HA
	12.5	7.34	6838	0.007	LA > HA
β	13.3	8.59	6838	0.003	LA > HA
	17.2	7.03	6838	0.008	LA > HA
	18.8	47.50	6838	0.001	LA > HA
	21.1	10.64	6838	0.001	LA > HA
	22.7	8.59	6838	0.003	HA > LA
	24.2	3.65	6838	0.001	HA > LA
	25	9.33	6838	0.002	HA > LA
	26.6	20.36	6838	0.001	LA > HA
	28.1	11.51	6838	0.001	HA > LA
	29.7	26.19	6838	0.001	HA > LA

Significant differences at $p < 0.05$ with Tukey's HSD correction.

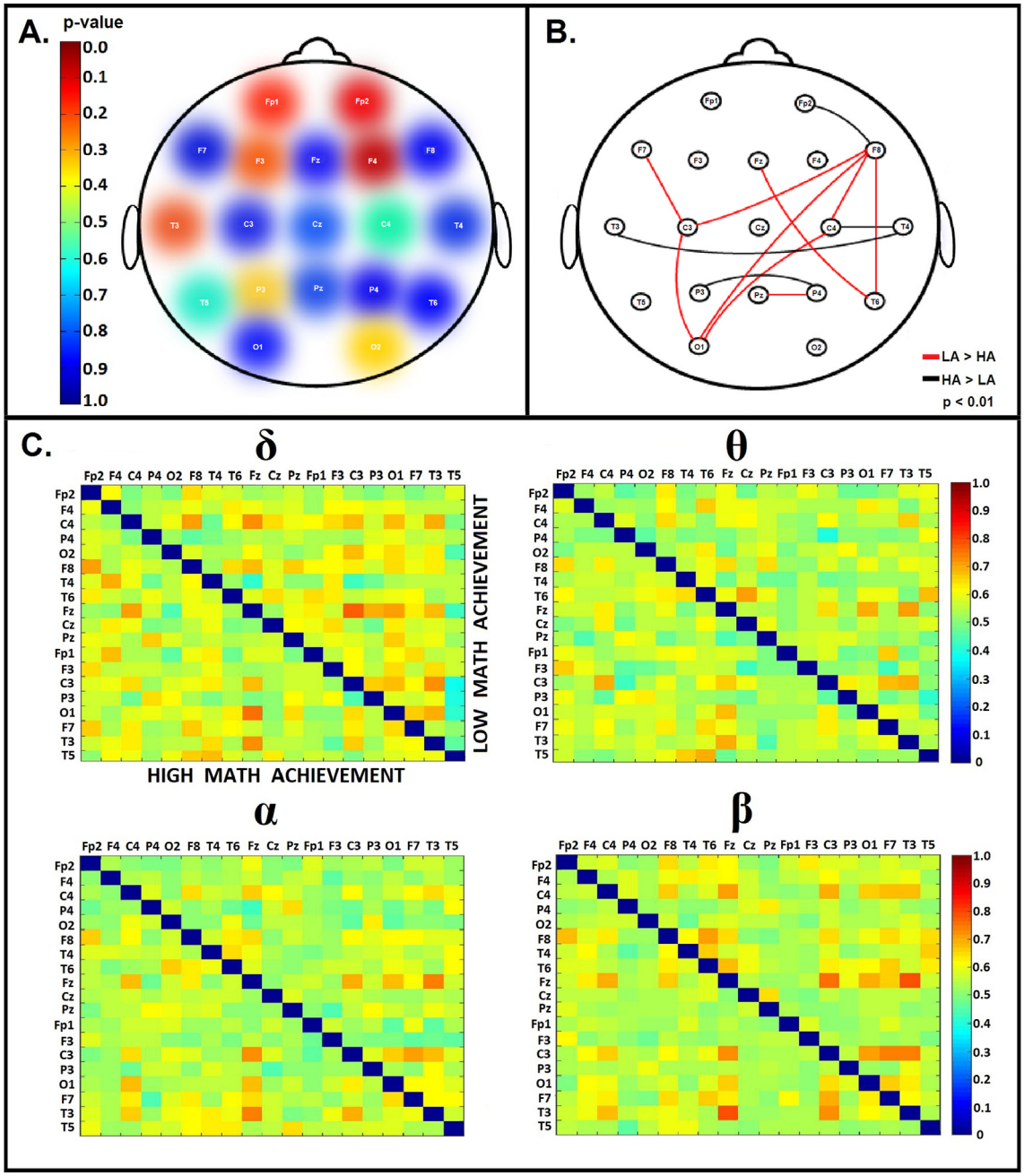

Fig. 1. The left upper section of the figure represents the individual scalp locations with significantly different coherence-based connectivity coefficients between groups. A connectivity graph representing the most statistical significant ($p < 0.01$) coherence relationships in both groups appears at the right upper section (LA > HA in red, HA > LA in black). The bottom of the figure shows the coherence matrices obtained for delta, theta, alpha, and beta frequency bands in both groups.

coherence group differences that involve frequencies from almost all the traditional frequency bands, although most within the beta spectrum. Interestingly, LA showed higher coherence in theta and alpha bands, while it was significantly greater for HA in the beta band, particularly at higher beta frequencies.

The analysis also identified the scalp locations in which greater coherence differences were obtained between the groups. Despite the extensive connectivity pattern shown, the more significantly coherence values were reached by recording sites predominantly distributed over the right hemisphere, regardless of group membership. The left section of Fig. 1 shows the scalp locations with significant between-group coherence differences.

Moreover, analysis of the specific connectivity between the scalp locations, marked as those with greater coherence differences between the groups, shows a distinctive pattern for LA and HA in the resultant graph (the right section in Fig. 1 shows the connectivity graphs in both groups). As it is depicted in Fig. 1, the LA-group has a more extended coherence topographic pattern that mainly involves frontoparietal areas, predominantly affecting the right hemisphere, while comparatively, a more bilateral and predominantly parietal connectivity is observed in the HA group. The bottom part of the Fig. 1 illustrates a comparison of the coherence matrices obtained from both groups in delta, theta, alpha, and beta frequency bands.

Fig. 2 compares the changes in the absolute power of the two groups of children across the time–frequency domain at parietal locations. Most of the visible changes across this domain predominantly occurred

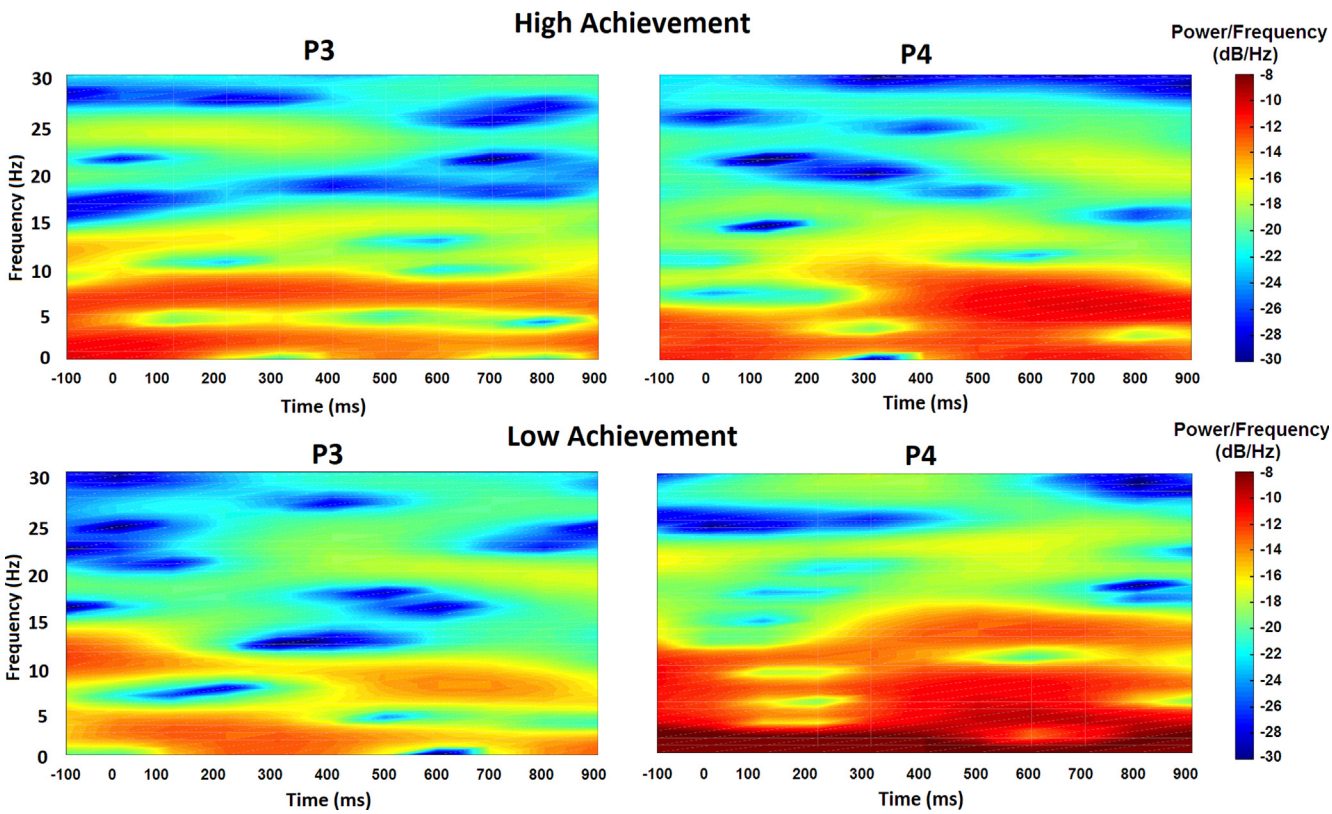

Fig. 2. The spectral power (dB/Hz) of the parietal locations in children with different arithmetic achievements while performing a symbolic magnitude comparison task is shown in time–frequency distribution illustrations.

over the right hemisphere and in the time period between 300 and 700 ms. HA children showed an earlier and sustained increase in the power of the beta band, followed by a later increment in theta and delta bands, while LA showed an earlier and persistent increment of the power in delta, theta and alpha bands, with minor parallel changes on higher frequencies. In addition, changes in the estimated power seemed to be greater at the right in LA group when compared to those observed in HA.

4. Discussion

As expected, LA's behavioral performance was more inaccurate than that of HA, confirming previous reports about the relationship observed between math difficulties and the processing of numerical magnitudes (De Smedt and Gilmore, 2011; Iuculano et al., 2008; Rousselle and Noël, 2007). In addition, the "distance effect" which has been sufficiently supported by several studies on numerical processing (e.g., Banks and Flora, 1977; Moyer and Landauer, 1967; Schwarz and Heinze, 1998), can be observed in the results from both groups. However, the distance effect was not significantly different between the groups, probably because the relationship between discriminability and size-ratio is modulated by task demands and familiarity with the compared stimuli (Leibovich, Ashkenazi, Rubinsten, & Henik, 2013). The problem size was originally conceived as a variable to be balanced by the present experimental design. Therefore, an exhaustive analysis of the interaction between arithmetic achievement and problem size was beyond the scope of this research, thus representing a limitation to be further investigated in future studies.

LA children showed significantly lower IQ compared to their peers. It is not surprising because dyscalculia has been found as comorbid in children with lower IQ levels (2005; Shalev, Manor, Auerbach, & Gross-Tsur, 1998). However, it is important to emphasize that none of the participants had an IQ score lower than 90, which corresponds to an average level.

It has been largely debated whether numerical processing operates within a specialized brain module or if this type of processing runs in a more generalized system used to process both numerical and non-numerical magnitudes. Several studies have evaluated comparative judgments on different kinds of numerical and non-numerical magnitudes, and they have reported both distinct and overlapping neural populations (e.g., Cohen, Lammertyn, & Izard, 2008; Dormal, Dormal, Joassin, & Pesenti, 2012; Fias, Lammertyn, Reynvoet, Dupont, & Orban, 2003; Pinel, Piazza, Le Bihan, & Dehaene, 2004). In this regard, a recent quantitative meta-analytic study seems to support the hypothesis that regions along the parietal and frontal cortex host a general magnitude processing system for numerical magnitude processing (See Sokolowski, Fias, Mousa, and Ansari (2017b), for a comprehensive review).

A greater involvement of frontal regions during task performance was found in the LA group. This finding is consistent with behavioral studies showing slower development of number-related competencies in LA and learning-disabled children, along with working memory deficits (Arsalidou and Taylor, 2011). It also coincides with behavioral studies of group differences specifically, delays in fluency of number processing and slower development of number-related competencies in LA and learning-disabled children and more working memory deficits in these children (Geary, Hoard, Nugent, & Bailey, 2012).

In general, low EEG coherence between frontal and parietal regions has been associated with poor executive performance (Teramoto et al., 2016). Moreover, it has been reported that frontal coherence in delta and beta frequencies positively correlated with memory and executive function performance (Fleck, Kuti, Brown, Mahon, & Gayda-Chelder, 2016). However, brain imaging studies have revealed that functional activation associated with numerical task performance is not restricted to frontal and parietal cortices, but also includes regions related with the dorsal and ventral visual pathways, as well as sub-cortical structures

(Arsalidou and Taylor, 2011; Kaufmann, Wood, Rubinsten, & Henik, 2011).

From this angle, greater frontal involvement in LA children might be interpreted as an effort to recruit additional cognitive resources to fulfill task demands. This could be because frontal brain regions seem to host the magnitude representations of the numbers (see Sokolowski, Fias, Bosah Ononye, and Ansari (2017a), for an extensive review), but they are also involved in numerical tasks (Arsalidou and Taylor, 2011). Considering that there is an ontogenetic specialization of the left inferior parietal, and the lateral occipito-temporal cortices for mental arithmetic and the processing of symbolic visual stimuli, respectively (see Ansari (2008), for an extensive review), the greater frontal participation in LA might represent an overexpression of the fronto-parietal developmental dynamic, probably denoting a lower specialization of the numerical processing. However, this interpretation should be taken cautiously, due to the limited spatial resolution of the EEG analysis tools.

In contrast to what was observed in LA, the more focalized coherence found in HA over parietal areas might reflect the numerical processing *per se*, probably suggesting a higher degree of automation or system specialization in this group of children with better arithmetic skills.

Previous studies regarding EEG spectral parameters have shown the involvement of theta, alpha and delta bands in arithmetic processing and developmental changes therein. Recently, Hesse, Schmitt, Klingenhoefer, and Bremmer (2017) described a bilateral increase in evoked theta power while performing a visual task in which different numerosity was presented as task-irrelevant stimuli. These variations in theta band were interpreted as evidence of a pre-attentive processing of numerical visual information. Moreover, the dynamic change of attentional demand during information processing has been related to alpha activity (Wei, Zhao, Yan, Duan, & Li, 1998). In this context, the differences observed in the temporal oscillations of these EEG bands suggest that arithmetic skills seem to mainly affect the processing phases subsequent to encoding. This notion seems to agree with the findings of Thevenot, Fanget, and Fayol (2007), who developed an operand recognition paradigm for studying the strategies used by individuals to solve arithmetic problems, and found that high-skilled individuals relied preferably on retrieval strategies while less-skilled individuals resort to nonretrieval strategies.

There is sufficient evidence suggesting a direct association between numerical processing and changes in other EEG frequency bands, however the involvement of the beta band is still poorly known. Indeed, the presented results show that most of the spectral coherence differences between the groups occurred in the beta band hence suggesting a major role of this EEG band in distinguishing among children based on arithmetic numerical abilities. In this regard, previous reports on changes in beta activity during mathematical problem solving (Lin, Jung, Wu, She, & Jung, 2015; Mizuhara et al., 2005), arithmetic tasks (Widagdo, Pierson, & Helme, 1998), and during the learning process on how to solve proportional reasoning tasks (Vecchiato et al., 2013) suggest the need for further exploration of the association between the beta band spectrum and math processing skills.

Interestingly, HA showed greater coherence on most of the upper beta power frequencies (20–30 Hz), which have been associated with successful integration over central and parietal electrodes, probably reflecting general control processes that can facilitate temporal integration (Geerligs and Akyürek, 2012). This greater coherence might also be interpreted as part as a more efficient attentional top-down processing in which beta band oscillations may relate not only to the involvement of top-down processing but also to the content of the top-down signal (see Engel and Fries, 2010, for review).

It is also important to highlight that the differences found between the groups were not restricted to the beta band. Given that EEG coherence represents a solid measure of functional connectivity in the human cortex, the present results suggest that symbolic numerical comparisons in children may involve a more complex functional integration of several brain networks, in which math achievements might be related with specific dynamic functional interconnections.

4.1. Limitations of the present study

The present study has three main limitations. The first one concerning the restricted number of artifact-free EEG epochs included in the analysis, and its impact on the electrophysiological expression of the evaluated processes. The second concerning the lack of direct functional evidence from the neuroanatomical sources involved in the numeric processing, thus restricting the theoretical interpretation of the results, which is a topic that should be addressed in future studies. And finally the third, that we were unable to apply a realistic head geometry model that would adjust to the exact age of our sample even though this would have been ideal.

5. Conclusions

Middle childhood differences in mathematical achievement are illustrated by EEG coherence analyses and underlying brain functional dynamic while comparing symbolic magnitudes. The present results suggest that different math skills represent dissimilar degrees of a number-processing specialization system that probably involves several complex interacting neural networks with a distinct topographic distribution; mainly affecting the processing steps subsequent to the encoding phase and probably encouraging the choice of the cognitive strategy more suitable to the available processing resources. Nevertheless, further studies are necessary to elucidate if numerical comparison task-related functional connectivity is directly associated with domain-general factors that participate in mathematical learning, such as, working memory, reasoning, and processing speed, among others. We also believe it is essential that future research aims to clarify the seemingly important role of the EEG beta band in this particular type of cognitive processes.

6. *Conflicts of interest*

None.

7. *Funding*

Universidad de Guadalajara.

Appendix A. Supplementary material

Supplementary data associated with this article can be found, in the online version, at https://doi.org/10.1016/j.bandc.2018.04.006.

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